

Rajesh Majumder

PhD Researcher | Biostatistician

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Technical Skills

- Study Designs and sample size calculation
- Strong foundation in statistical inference and advanced analytical modeling
- Proficient in developing clear, insightful data visualizations
- Skilled in modern missing-data methodologies and imputation strategies
- Experienced with resampling techniques, including bootstrap and permutation methods
- Proficient in longitudinal and repeated-measures data analysis
- Advanced regression expertise, including Ridge, LASSO, GAM, GLMM, and GAMLSS.
- Solid command of advanced multivariate statistical methods and Machine Learning techniques.

Programming Skills



About Me

Biostatistician and PhD researcher with over four years of experience in clinical, epidemiological, nutritional, and population health research. Primary research focuses on child growth, nutritional epidemiology, anthropometric growth references, obesity-related measures, and cardiometabolic risk assessment using large-scale health datasets. Experienced in analysing nationally representative health surveys and collaborating with multidisciplinary teams from organisations including NIN, ICMR, the World Bank, and the Health Effects Institute (HEI). Research contributions span child health, nutrition, disease burden estimation, and environmental epidemiology, including assessment of air pollution exposure and cardiovascular health outcomes.

Experience

Project Research Scientist II (Non – Medical) [May,2026 - present]

Project: **ICMR Burden of Disease IIT-B**
Institute: **St. John’s Research Institute, Bangalore, India**

Roles & Responsibilities

- Contribute to the development of district-level estimates of diabetes burden across India using large-scale epidemiological datasets.
- Perform statistical analysis, data integration, and modelling of diabetes prevalence and related health outcomes.
- Apply advanced epidemiological and statistical methods for disease burden estimation and risk assessment.
- Develop reproducible analytical workflows for processing, analysing, and visualizing population health data.
- Support generation of evidence for identifying regional disparities and informing public health strategies for diabetes prevention and control.

Research Assistant (Statistician) [Feb,2022 - present]

Division: **Biostatistics and Epidemiology**
Institute: **St. John’s Research Institute, Bangalore, India**

Roles & Responsibilities

- Designed statistical analysis plans for national and international research studies.
- Performed data management, cleaning, transformation, and quality assessment.
- Developed regression, generalized additive, mixed-effects, and distributional models.
- Applied causal inference methods and sensitivity analyses.
- Conducted sample size estimation and statistical power calculations.
- Prepared publication-ready statistical reports and scientific outputs.
- Provided statistical mentoring and training to researchers and students.

Education

Ph.D. Scholar [2024-Present]

Manipal Academy of Higher Education (MAHE), Manipal, India
Division of Biostatistics and Epidemiology, St. John’s Research Institute, Bengaluru, India (a unit of CBCI Society for Medical Education)

M.Sc. In Statistics [2019-21]

West Bengal State University
With 8.68 CGPA

B.Sc. In Statistics [2016-19]

Acharya Prafulla Chandra College
With 63% marks

Teaching and Mentoring Skills

- Mentoring students and interns in biostatistics, epidemiological methods, statistical modelling, and research data analysis.
- Delivering practical training in R programming, statistical computing, data visualization, and reproducible research workflows.
- Supporting multidisciplinary research teams in study design, statistical analysis, and interpretation of findings.

Awards And Recognition

- Third Best Poster Presentation Award in 5th Annual Research Day, 2025, Organized by St. John’s National Academy of Health Sciences.
- Best Oral Presentation Award in ISMSCON 2023.
- Best Oral Presentation Award in ISMSCON 2022.
- First Rank in M.Sc. Statistics.

Certifications and Workshops

- Machine Learning, Data Science and Deep Learning with Python – Udemy.
- Introduction to SQL – Simplilearn.
- National Workshop on Randomized Controlled Trials, Kannur University in collaboration with McMaster University, Canada.
- Workshop on Geospatial and Environmental Exposure Tools for Evidence Synthesis, ICMR-NCDIR and CCDC.

Personal Interests and Activities

Music enthusiast with interests in singing, guitar and harmonica performance.

References

Available upon request.

Publications

- Ghosh, S., Majumder, R., Sachdev, H.S. et al. Customization of WHO Under-five Growth Standards for an Appropriate Quantification of Public Health Burden of Growth Faltering in India. Indian Pediatr 60, 804–810 (2023). <https://doi.org/10.1007/s13312-023-3009-8>
- Majumder, R., Kurpad, A.V., Sachdev, H.S. et al. Anthropometric Growth Reference for Indian Children and Adolescents. Indian Pediatr 61, 425–434 (2024). <https://doi.org/10.1007/s13312-024-3128-x>
- Krishna, S.R., Muzumder, S., Johnson, S. et al. Predictive Factors for Esophageal Stricture Following Definitive Chemoradiotherapy in Esophageal Carcinoma. J Gastrointest Canc 56, 65 (2025). <https://doi.org/10.1007/s12029-025-01192-1>
- Pullakhandam R, Augustine LF, Banjara SK, Dasi T, Palika R, Majumder R, et al. Impact evaluation of a population-based ‘Screen and Treat for Anaemia Reduction (STAR)’ strategy: a cluster randomised trial in rural Telangana, India. BMJ Global Health. 2025;10:e020633. <https://doi.org/10.1136/bmjgh-2025-020633>
- Majumder R, Kurpad AV, Sachdev HS, Kuriyan R, Thomas T, Ghosh S. Untangling weight from height: A generalised body mass index (gBMI) approach to evaluating child and adolescent nutritional status in India. Indian J Med Res. 2026;163:655-63. doi: 10.25259/IJMR_2555_2025.

Methodological Research Project

Performance of LASSO when one or more covariate is/are Missing Not at Random (MNAR)

Motivation

- Investigated variable selection under MNAR mechanisms with multicollinearity.
- Developed an alternative inverse probability weighted LASSO framework.
- Implemented LARS-based optimization approaches using R.

Research Software and Analytical Tools Developed

- **StudySizer: A Sample Size Calculator**

Link: <https://studysizer.streamlit.app/>

Motivation: The tool designed in Python, to empower researchers, healthcare professionals, and students with statistically sound sample size calculations for various study designs including estimation, testing, regression, and reliability.

- **Anthropometric Z Score Calculator for Indian 0 to 19 Years children**

Link: <https://india19azcal.streamlit.app/>

Motivation: An interactive tool designed in Python, based on India’s national growth reference that calculates Z-scores, visualizes growth centiles, and assesses growth faltering risk for children and adolescents aged 0–19 using single or bulk data inputs.

- **Habitual/Usual Intake Calculator**

Link: <https://smith123.shinyapps.io/USIC/>

Motivation: A tool designed in R shiny to estimate individuals’ habitual or usual nutrient intake by accounting for daily dietary variation, using both the standard NRC approach and advanced methods such as Gamma regression and mixture distribution models.

Future Research Interests

My future research interests include investigating the interplay between biological, environmental, and social determinants of child health. I am particularly interested in integrating epidemiological, genetic, and molecular data to understand how early-life exposures, including environmental factors and gut microbiome-related pathways, influence growth, nutrition, and long-term health outcomes.